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Predicting Sales for ABC grocery company using Python

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Contents

1. Introduction 2
2. Data Description 2
3. Data Preprocessing 4
4. Extensive Exploratory Data Analysis (EDA) 5
5. Project Flow Chart 7
6. Segmentation and Time Series Analysis 8
6.1. Grouping Solely by Product 8
6.2. Grouping Solely by Store 9
6.3. Grouping by Both Store and Product 9
7. Model Selection 9
7.1. Linear Regression 9
7.2. Random Forest 10
7.3. Gradient Boosting Regression 10
7.4. MLP Regressor 11
7.5. LightGBM 11
7.6. XGBoost 12
8. Feature Selection/Engineering 12
9. Data Splitting 13
9.1. Predefined Split Cross-Validator 13
10. Hyperparameter Optimization using GridSearchCV 13
11. Prediction Process 14
12. Performance Evaluation 14
12.1. Model Assessment Metrics 14
12.2. Overfitting and Data Leakage Examination 15
12.3. Insights and Model Selection Challenges 15
13. Results 15
14. References 18
15. Appendix 19

1. Introduction:

Grocery retailers globally encounter a significant dilemma in managing the equilibrium between inventory levels and sales. Inaccurate sales predictions can result in challenges, such as excessive inventory with perishable goods or rapid depletion of popular items, leading to distressed customers and potential revenue loss (Pophali, 2023). Retailers need to exercise extra caution and organisation when expanding into new locations and introducing new products. This is crucial due to the unpredictability of demographics, consumer preferences, and seasonal trends impacting product sales.

Numerous supermarkets currently struggle with accurately forecasting their annual sales. This challenge primarily stems from a shortage of skills, resources, and expertise for practical sales estimation. In many cases, supermarket chain stores employ ad hoc tools and processes to analyse and project sales for the upcoming year. The reliance on traditional statistical methods for supermarket sales forecasting has left several challenges unattended, often leading to the development of predictive models that exhibit poor performance (Odegua, 2020).

We have used their dataset to model and efficiently resolve a problem faced by the ABC grocery retailer company in South America. Our strategy focuses on improving the precision of their sales predictions, reducing their dependency on subjective forecasting techniques, and overcoming the limitations of data support and lack of execution through automation. Our efforts have been directed towards achieving greater efficiency and accuracy in their sales predictions.

In this project, we have predicted sales for July 31, 2017, and August 15, 2017, from the given dataset using Python programming language and supervised machine learning models, mainly XGBoost and LightGBM (Chavali et al., 2023).

2. Data Description:

The dataset ‘Products_Information.zip’, provided by the company ABC, is a source of sales data covering a significant period from ***January 1, 2013, to August 15, 2017***. It offers insights into the sales of 33 product types across 54 stores, making it a crucial tool for understanding the company's performance over time. The dataset contains five key components essential for analysis: Date, Store Number, Product Type, Sales, and Special Offer.

Date: The date field in the dataset is fundamental, indicating the chronological order of each record. Spanning from the beginning of 2013 to mid-August 2017, this temporal information is vital for examining trends, seasonality, and the overall progression of sales over the provided time frame.

Store Number: Each record is associated with a specific store, identified by its indicator number (store_nbr). With 54 distinct store numbers, the dataset provides a detailed breakdown of sales performance across various retail locations.
Product Type: Covering 33 product types, the dataset categorises products. This breakdown enables a detailed analysis of the performance of each product type, offering insights into which products are more successful and may require additional attention or marketing efforts.
Sales: The 'sales' field represents the total sales for a specific product type on a given date. This quantitative measure is pivotal for evaluating the financial performance of the company. Analysing sales data over time can reveal patterns, fluctuations, and the impact of external factors on revenue generation.
Special Offer: The 'special_offer' field provides information on the strength of promotional efforts associated with each record. A numerical value is assigned, with a higher number indicating a more intense promotional campaign. Understanding the correlation between special offers and sales can unveil the effectiveness of promotional strategies and guide future marketing endeavours.

In the early stages of analysing the dataset, we began by loading and examining the data using various libraries, including pandas and NumPy, as well as seaborn and matplotlib.pyplot for visualisation. These tools enabled us to navigate the dataset easily and better understand its structure and contents. We used pandas' info () and head () functions to assess the dataset's dimensions and variable composition.

To ensure the quality and usability of the dataset for future analysis, we addressed any missing value issues. Surprisingly, our dataset lacked missing values in all columns, making imputation techniques unnecessary. This peculiar feature simplified our workflow while boosting the accuracy of our findings.

Furthermore, we optimised the dataset by transforming the 'date' column into a DateTime format and setting it as the index of the DataFrame. This deliberate action improves our ability to analyse data based on specific dates, allowing for more detailed and temporally informed insights throughout our extensive examination of the product dataset.

id	date	store_nbr	product_type	sales	special_offer
0	2013/1/1	1	AUTOMOTIVE	0	0
1	2013/1/1	1	BABY CARE	0	0
2	2013/1/1	1	BEAUTY	0	0
3	2013/1/1	1	BEVERAGES	0	0
4	2013/1/1	1	BOOKS	0	0
5	2013/1/1	1	BREAD/BAKERY	0	0

Figure 1: Head of data set

To summarise, the dataset named 'Products_Information.zip' plays a vital role in predictive analytics. Its significance lies in its ability to provide valuable information for predicting sales across different product types and stores over 16 days. This dataset is a crucial foundation for constructing predictive models using temporal details, store numbers, product types, sales figures, and insights into promotional efforts. Ultimately, this dataset serves as a strategic tool for refining predictive capabilities, enabling company ABC to make well-informed decisions, optimise inventory management, and improve the accuracy of sales forecasting overall.

3. Data Preprocessing

Data preprocessing is a crucial step in preparing our dataset for analysis and model training. This involves handling missing values, transforming data, and ensuring its compatibility for further processing.

Handling Missing Values: To ensure the quality of our dataset, we began by checking for missing values using the 'isnull().sum()' function to count the missing values in each column. Fortunately, the analysis revealed that there are no missing values in the dataset, eliminating the need for imputation or other strategies to address missing data.
Data Transformation (Date Column): One important transformation involved converting the 'date' column to datetime format using the 'pd.to_datetime()' function. This conversion allows for better temporal data handling, enabling us to extract valuable insights based on dates.
Check for Updated Processing: After handling missing values and transforming the date column, we performed a comprehensive check to ensure that all processing steps were accurately implemented. Finally, we saved the processed data into a CSV file for future reference and ease of access, the screenshot of which can be seen in Figure 2.

```
In [11]: print(product.head())
            id  store_nbr product_type  sales  special_offer
date
2013-01-01   0          1   AUTOMOTIVE    0.0              0
2013-01-01   1          1    BABY CARE    0.0              0
2013-01-01   2          1       BEAUTY    0.0              0
2013-01-01   3          1    BEVERAGES    0.0              0
2013-01-01   4          1        BOOKS    0.0              0
```

Figure 2

4. Extensive Exploratory Data Analysis (EDA):

Sales Trends Analysis: Aggregate the sales in all 54 stores, including 33 kinds of products, and plot overall sales trends over time to understand the general sales behaviour and identify any apparent patterns or anomalies.

Figure 3: Overall sales over time

Sales Distribution and Comparison: Analyze the sales distribution using histograms and compare sales across product types with box plots. This helps us to identify outliers, variations, and other vital insights. In Figure 4, we can observe that several products have almost zero sales overall. On the other hand, there are a few relatively high sales values in the Beverage, Grocery I, and Produce category.

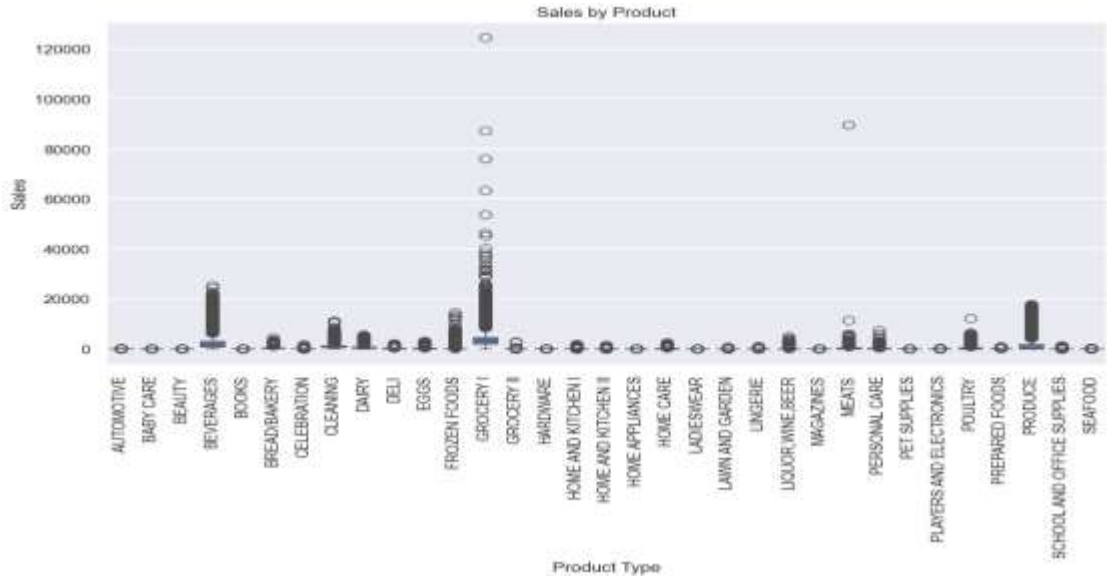

Figure 4: Sales vs. Product type

Average Sales of Stores and Products Combinations: The mean sales numbers across each combination of store number and product types have a considerably wide range (Figure 5). Acknowledging this fact, we assumed that each combination should have suitable models for forecasting.

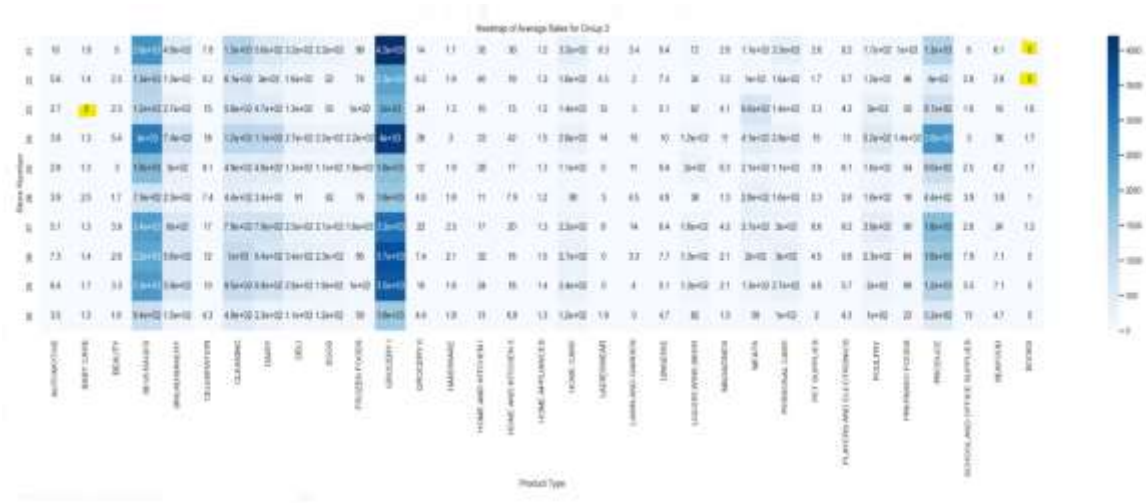

Figure 5: Average Sales heatmap of Stores and Product combinations

Zero and Non-zero sales combinations:

According to the pie chart in Figure 6, we discover that there are some combinations that contain zero sales. To ensure that we process the data correctly, we have to separate the data that has zero sales throughout the time and focus on non-zero sales.

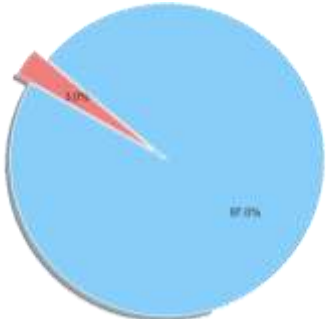

Figure 6: Pie chart

Statistical Summary: Provide a statistical summary of sales, such as mean and standard deviation, to understand sales data's central tendency and dispersion. The mean sales value in all stores and all products is about 358 units, with a minimum value of 0 and a maximum value of 124717 units.

5. Project Flow Chart

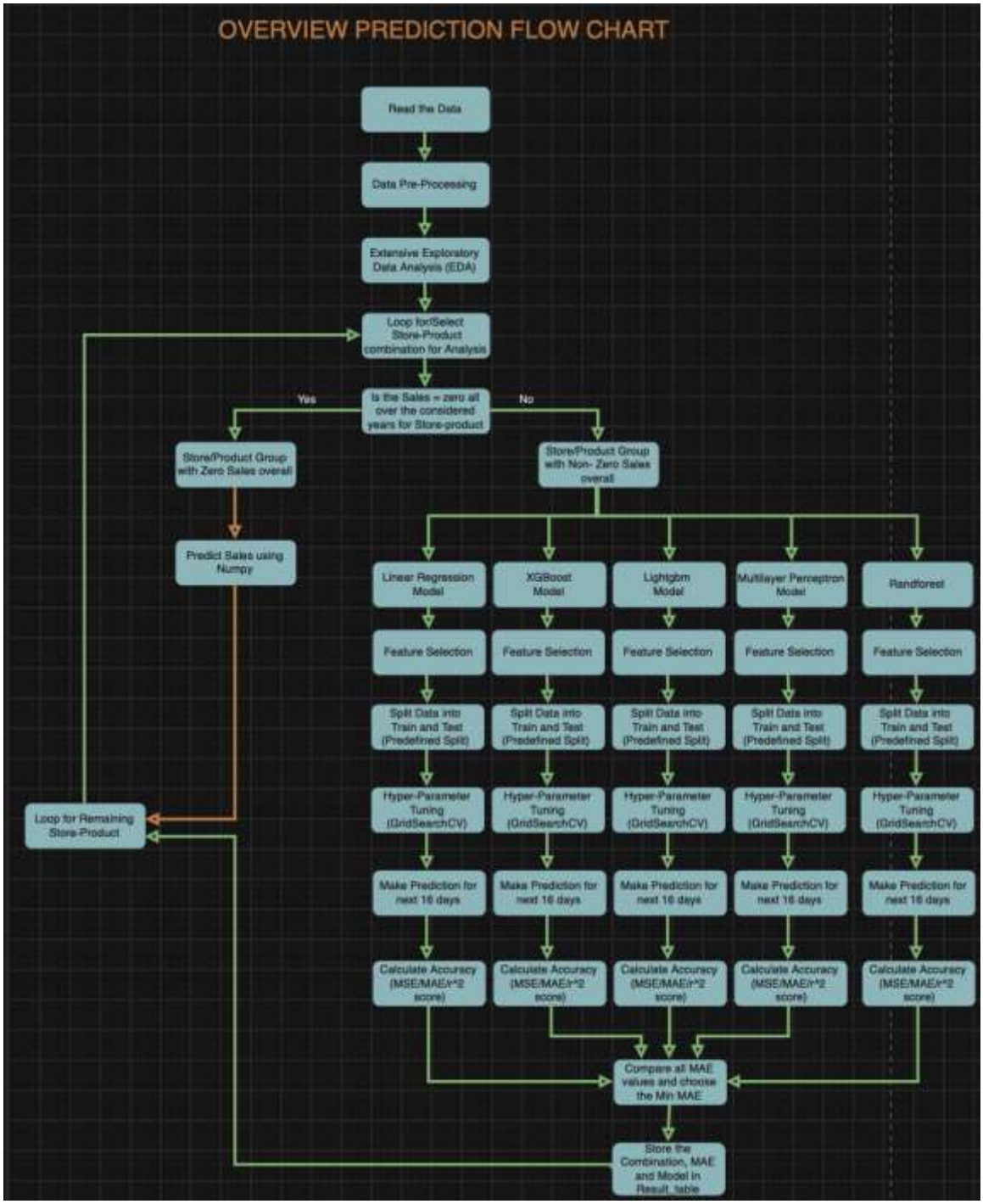

Figure 7

It is clear from Figure 7 that the first step in our analysis is to read and import the data. Following this, we use data preprocessing to ensure our dataset is clean, consistent, and ready for analysis.

We performed exploratory data analysis (EDA) on our dataset to gain insights. Based on the analysis, we selected specific combinations of stores and products to focus on for targeted predictions. Our predictive journey diverges from here. For accurate predictions when sales are zero, we use NumPy methods. On the other hand, for non-zero sales, we rely on the power of five different models: Random Forest, Linear Regression, MLP, XGBoost, and LightGBM.

Next, we perform feature selection to optimize the performance of our model. Next, we split our dataset into training and testing sets to prepare for model evaluation. To improve the accuracy of our models, we use hyperparameter tuning, fine-tuning the parameters for each model. The ultimate goal is to maximize predictive accuracy. We then calculate the Mean Absolute Error (MAE) for each model and compare the MAE values of all models. This helps us to identify the model with the minimum MAE, which is the optimal model. We store this model in a results table, capturing the essence of our predictive prowess. As we loop through these steps for each store, this iterative process ensures a comprehensive analysis of our entire dataset.

6. Segmentation and Time Series Analysis:

Creating Segments: Segment the data by unique combinations of store numbers and product types, storing each segment in a dictionary for easy access and individual analysis.
Visual Time Series Analysis: Plot individual time series data for selected store-product combinations to visually inspect trends, seasonality, and other time series characteristics.
Statistical Testing for Stationarity: Perform the Augmented Dickey-Fuller test on each time series segment to check for stationarity. Segments are categorised based on whether they exhibit zero sales, non-stationary, or stationary characteristics. The purpose of implementing this strategy is to build an appropriate forecasting model due to the investigation of time series features from our data set.
Eventually, I decided only to split the zero-sales and non-zero-sales sub-data set.

6.1. Grouping Solely by Product

Grouping data exclusively by-products allowed us to conduct an in-depth product performance analysis. The company can use this grouping to define marketing and production strategies for each product. Moreover, focusing solely on products can help predict overall product demand, regardless of the specific store.
However, this method has some significant shortcomings. By focusing only on products, the model lacks store-specific dynamics, potentially limiting the effectiveness of the prediction model.

6.2. Grouping Solely by Store

We also considered grouping our data solely by stores. The main key advantage is its simplified analysis process, as focusing on the store level minimises the complexity associated with analysing data across multiple products. This approach proves effective for evaluating the overall performance of each store, making it easier for companies to make strategic decisions regarding product allocation.

On the other hand, the primary limitation is the loss of insights into our products, as the analysis does not consider how each product performs within each store. Although grouping only by store provides a good overview of store performance, it is insufficient for companies looking for detailed insights into their product dynamics.

6.3. Grouping by Both Store and Product

In terms of data, we first tried grouping data by both store and product. This approach showed insights into our dataset of sales patterns for each product within each store, providing a more accurate understanding of the performance of each product in each specific store.

However, this approach can be complex and intensive, especially when dealing with numerous stores and products. In addition, this introduces a higher risk of overfitting when creating predictive models, posing challenges in making reliable forecasts.

Despite these challenges, the benefits of obtaining explicit knowledge and the ability to develop strategies often exceed the disadvantages, making this approach valuable for companies to understand their sales dynamics better.

7. Model Selection:

7.1. Linear Regression:

Linear regression is a widely employed method particularly in scenarios where the prediction of effects can be measured and modelled against multiple input variables. It also serves as a data evaluation and modelling technique that establishes linear relationships between dependent and independent variables. This approach includes modelling relationships derived from the analysis and learning of current training results,

allowing for examining relations between dependent and independent variables (Maulud & Abdulazeez, 2020).

However, when applied to predict non-zero sales, linear regression exhibited suboptimal results and unsatisfactory performance. The limitations of linear regression in capturing the complexities of non-zero sales scenarios prompted us to explore alternative approaches. In response to the challenges faced with linear regression, we adopted a strategy of utilising multiple models. The need for faster computation and an improved distribution of the computation load across different models drove this decision. By diversifying our modelling approach, we aimed to leverage the strengths of various models to better address the intricacies of predicting non-zero sales, ultimately enhancing our predictive modelling framework's overall performance and efficiency.

7.2. Random Forest:

We have chosen the Random Forest model because it stands out as a robust supervised machine-learning technique employed in regression and classification scenarios (Naik et al., 2022).

It operates by constructing an ensemble of decision trees during training, employing a strategy known as bagging, where each tree is trained on a different subset of the data. Additionally, random feature selection at each split ensures diversity among the trees. Random Forest uses the ensemble during prediction by combining the individual tree predictions. In regression, the final prediction is determined by averaging the predictions of all trees. This ensemble approach contributes to the model's accuracy and resilience, with the randomness introduced during training to mitigate overfitting. It can handle extensive data without much risk of overfitting. It is considered one of the best general-purpose models (Odegua, 2020).

7.3. Gradient Boosting Regression:

After trying several different models and not seeing significant improvements in prediction, we figured to use Gradient Boosting Regression, combining many simple models to improve the mistake and finally create a more accurate prediction (Zhang & Haghani, 2015).

The primary considerations of selecting the model are:

High accuracy: Gradient Boosting is an ensemble of weak learners; it resamples the training data after conducting each model to improve the overall performance of the prediction (Schapire, 1990).

While using Gradient Boosting Regression indeed improved the model performance, there were several concerns:

Computationally Intensive & Time Consuming: The dataset provided is exceptionally large; building the Gradient Boosting model by trying multiple different models may require more time and computer resources. Optimising hyperparameters in Gradient Boosting, such as learning rate, number of estimators, and maximum depth through grid search or random search, also required computational resources and time.
Overfitting: Gradient Boosting builds complex models by combining multiple weak learners. That leads to the risk of over-correcting the errors from each model, potentially resulting in low errors in training data but high errors in testing data (Natekin & Knoll, 2013).

7.4. MLP Regressor:

In the process of selecting an appropriate machine learning model for predicting sales for ABC retailers, the Multilayer Perceptron (MLP) Regressor was chosen. The primary considerations that made us consider MLP Regressor for the regression task were, firstly, the ability to model complex relationships, allowing for more accurate predictions. Secondly, the regressor's flexibility with numerical and categorical features provided a broad range of applications in regression analysis. Lastly, through its ability to learn the relationships of the input data, the Multilayer Perceptron (MLP) Regressor can uncover patterns that may not be easily identified through other methods.

Despite the Multilayer Perceptron (MLP) Regressor's improvement of our overall model for predicting sales, there were concerns about potential overfitting and the practicality of training MLP models, which can be time-consuming (Kigo et al., 2023).

7.5. LightGBM:

LightGBM is a novel Boosting framework model with two innovative technologies derived from traditional GBDT. The first, Gradient-basic one-side sampling technology (GOSS), strategically eliminates a substantial portion of data with a small Gradient, using only the remaining data to estimate information gain. This approach mitigates the influence of the long tail part of low Gradient. Given that data with large gradients are more crucial for information gain, the GOSS technique enables accurate estimates with significantly less data compared to traditional GBDT. The second technology, Exclusive Feature Bundling (EFB), efficiently reduces the number of features, decreasing the risk of overfitting. Additionally, LightGBM employs the histogram algorithm to replace traditional pre-sorted

feature points, reducing memory consumption and optimising the most time-consuming steps in GBDT (Wang, 2023).

In our modelling strategy, LightGBM played a pivotal role as one of the models used for predicting non-zero store-product combinations. Given the vast array of store-product combinations, we leveraged multiple machine-learning models, and LightGBM's capabilities were instrumental in providing accurate estimates and addressing concerns related to overfitting. This strategic integration significantly enhanced the overall efficiency of our predictive model, ensuring robust predictions in scenarios with diverse store-product combinations.

7.6. XGBoost:

As a part of our approach for predicting non-zero store product combinations, we integrated the Xgboost model. Solely relying on linear regression did not yield optimal predictions, as its performance was inefficient. Xgboost, with its encompassing linear scale-solving techniques and tree-learning algorithm, proved to be a valuable addition.

In contrast to the conventional GBDT, Xgboost introduced substantial enhancements, demonstrating increased efficiency. While the classical GBDT approach relies solely on first-order derivative information, Xgboost incorporates both GB-tree and a linear classifier. This allows it to perform linear regression or logistic regression with the L1+L2 penalty. Furthermore, Xgboost employs the second-order Taylor expansion of the loss function. It also introduces a regularisation term, augmenting the objective function to derive an optimal solution that balances the decline of the objective function and model complexity. This strategic addition helps prevent overfitting, enhancing the model's solving efficiency (Wang, 2023).

8. Feature Selection/Engineering:

Feature selection is a critical aspect of building effective predictive models, ensuring that only relevant and impactful variables are considered. In the context of our analysis for sales prediction, we evaluated various features and identified key considerations for inclusion or exclusion.

External Features: 'Id' and 'Store_nbr': These external features were deemed impractical for inclusion due to their limited impact on sales. Including them in the model would not significantly contribute to improving predictive accuracy.
Historic Sales Data: Solely relying on previous sales data yielded suboptimal performance for prediction. While historical sales data is undoubtedly important, depending solely on this feature did not provide the desired level of accuracy in our predictive models.

Lag Features: Lag features are created using the create_lag_features method with a specified number of lags (14 by default). We defined new lags for varying periods (e.g., 14, 21, 28 days).
Final selection of features: Combining historic sales data, lag values(for time), and information on special offers showed promising outcomes. This hybrid approach demonstrated improved predictive capabilities, suggesting that considering both aspects, along with lag values, enhances the model's ability to capture underlying patterns and make more accurate sales predictions.

9. Data Splitting

In training and validating our predictive model, a thoughtful approach to data splitting and cross-validation is paramount to ensure robustness and generalizability. We used a specific methodology to split our data into subsets for training, validation, and testing purposes. 80% of the dataset was allocated for training the predictive model, allowing it to learn patterns and relationships within the data. We selected various features through feature engineering like historic sales and special offers to use as parameters for the training process. 20% of the data was reserved for validation purposes, enabling us to assess the model's performance on unseen data and make adjustments accordingly. Finally, we selected a distinct timeframe from 31st July 2017 to 15th August 2017 for the test dataset to evaluate the model's predictive accuracy (scikit-learn, n.d.).

9.1. Predefined Split Cross-Validator

To ensure the sequential nature of training and validation data during the fine-tuning stage, we opted for the 'Predefined Split' function instead of traditional K-fold cross-validation. This approach allows us to maintain the temporal order of the data and ensures that the model is fine-tuned with training data before being validated on subsequent sequential data.

In summary, our integrated approach to data splitting, feature selection, and the use of the 'Predefined Split' function reflects a deliberate effort to create a robust and realistic evaluation environment for our predictive model. This approach considers both the temporal aspect of the data and the selected features crucial for accurate sales predictions (scikit-learn, n.d.).

10. Hyperparameter Optimization using GridSearchCV

Hyperparameter optimisation is crucial in fine-tuning machine learning models to achieve optimal performance. GridSearchCV, short for Grid Search Cross-Validation, is a systematic approach to exploring a predefined set of hyperparameter values for a given model (scikit-learn, n.d.). This method exhaustively evaluates all possible combinations

of hyperparameter values from a specified grid and identifies the set that produces the best model performance.

In our implementation, we utilised GridSearchCV to search for the optimal hyperparameters for our model. The key hyperparameters considered in the grid search varied based on the algorithm employed (scikit-learn, n.d.). For instance, if we were optimising hyperparameters for a Random Forest model, parameters such as the number of trees ('n_estimators'), the maximum depth of the trees ('max_depth'), and other relevant parameters would be included in the grid.

The grid search process involves fitting the model with each combination of hyperparameter values, performing cross-validation, and assessing the model's performance using a specified metric, such as accuracy or mean squared error. The set of hyperparameter values that results in the highest cross-validated performance score is then selected as the optimal configuration for the model.

11. Prediction Process

In our modelling process, Linear Regression, XGBoost, LightGBM, Random Forest, and MLP. This diverse ensemble of models was then applied to all store-product combinations to predict sales.

We calculated the Mean Absolute Error (MAE) value for each combination, a crucial metric for assessing prediction accuracy. The model with the least MAE for each store-product combination was selected, emphasising precision in the prediction outcomes.

To enhance clarity and representation, the chosen models and their corresponding MAE values were stored systematically in a data frame. This organised structure facilitates a comprehensive overview of the optimal models for each unique store-product combination.

12. Performance Evaluation

In the realm of forecasting models, evaluating their performance is crucial for ensuring accuracy and efficiency. To achieve this, we employed a diverse set of models, including Linear Regression, XGBoost, LightGBM, Random Forest, and MLP, along with NumPy zero methods for scenarios with no historical sales data.

12.1. Model Assessment Metrics

We assessed the accuracy and efficiency of each model using the Mean Absolute Error (MAE) metric. Reading the data and extracting meaningful patterns proved challenging,

especially when it came to classifying the data into relevant categories(zero and non zero). By applying these models to various store-product combinations, we aimed to identify the most effective model for each scenario by comparing the mean Absolute Error of each model and choosing the Model with min. MAE value . The MAE provided a quantitative measure of the prediction accuracy, guiding our selection process.

12.2. Overfitting and Data Leakage Examination

During our evaluation, we successfully mitigated overfitting concerns by implementing strategic measures in the preprocessing stage. Setting the 'date' column as the index ensured the models were trained and validated chronologically, preventing overfitting by maintaining the temporal order of the data. Moreover, thorough checks for null values using 'isnull().sum()' were conducted, addressing potential data inconsistencies and avoiding data leakage.

By adhering to best practices, such as setting the 'date' column as the index and carefully handling null values, we established a robust foundation for our models. These measures not only prevented overfitting but also ensured that data leakage risks were minimized. The reliability of our forecasting models was thus strengthened, contributing to their accuracy and effectiveness.

12.3. Insights and Model Selection Challenges

The process of selecting the most suitable models posed challenges, particularly in reading and classifying the data effectively. The intricacies of the dataset made model selection a nuanced task, requiring careful consideration of various factors to ensure optimal performance.

13. Results

In presenting the outcomes of our model's predictions, we utilised a visual representation to enhance clarity and understanding. In the below Figure 8, the y-axis is dedicated to average sales unit values, while the x-axis lists various models. For instance, if the linear regression model performed optimally for a specific store-product combination, its average sales would be grouped under the linear regression category. This approach visually demonstrated hidden patterns in model selection. Notably, if a combination's average sales unit fell within the range of 300 to 400, the linear regression model was identified as a highly suitable choice for predicting sales.

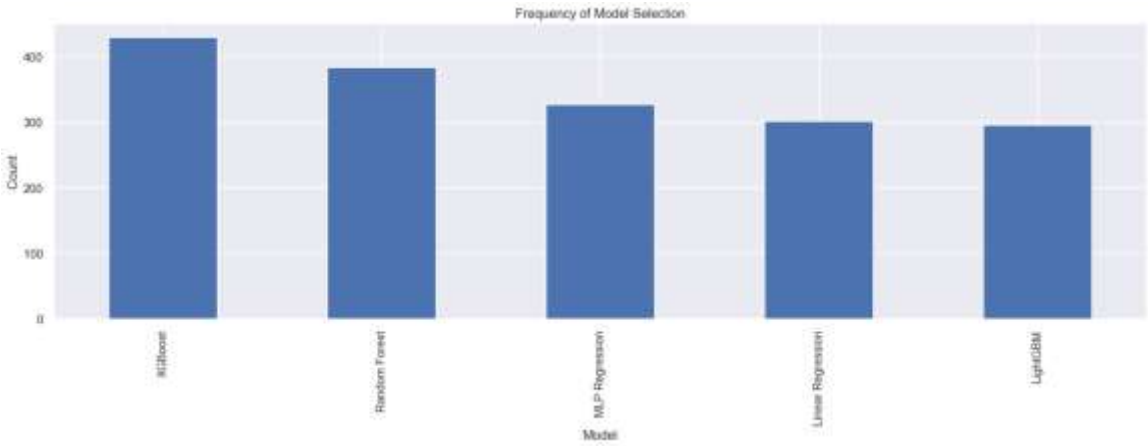

Figure 8

The below boxplot in Figure 9 illustrates the variability in sales ranges for predictions made by the best model for each combination. This further hinted at potential patterns in model selection. For instance, if a combination's sales range hovered around 1500, the linear regression model emerged as the most appropriate for sales prediction.

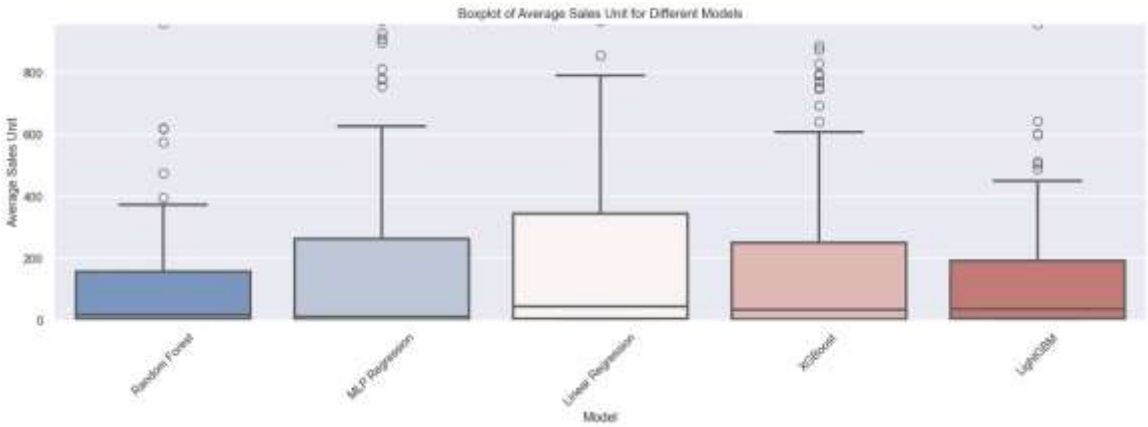

Figure 9

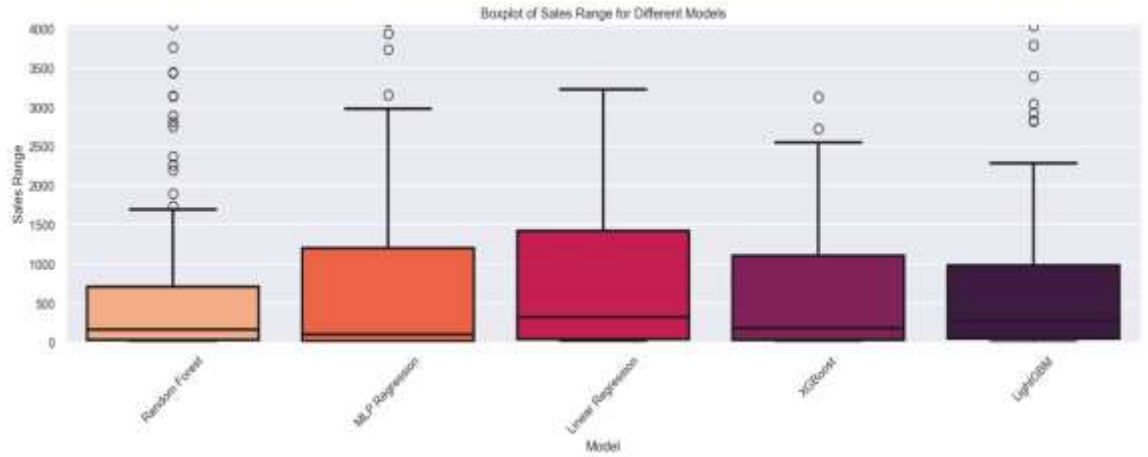

Figure 10

The comprehensive examination of the results in Figure 11, including the overall MAE distribution, revealed that almost 80% of predicted values had an MAE under 30.

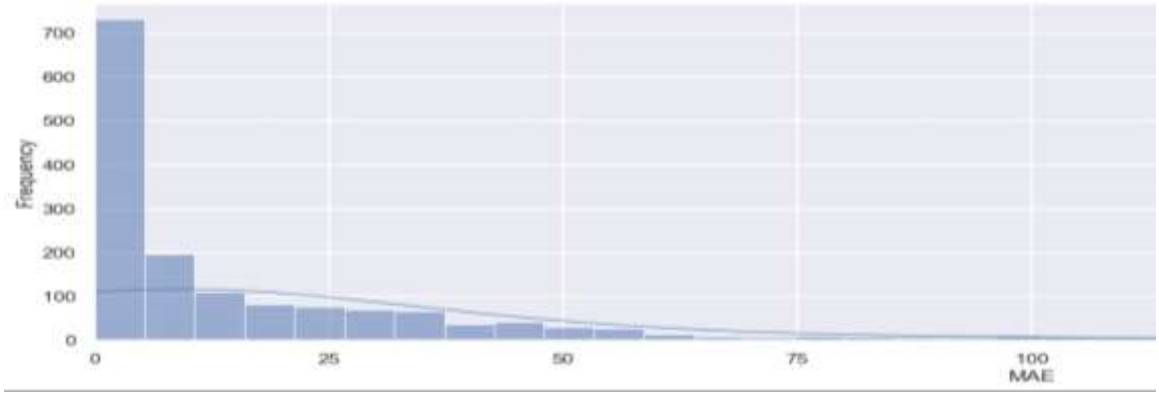

Figure 11

However, for a more accurate assessment, I opted for relative MAE , as shown in Figure 12, calculated as each store-product combination's MAE divided by their actual average sales. This provided a nuanced measure of accuracy tailored to each specific combination.

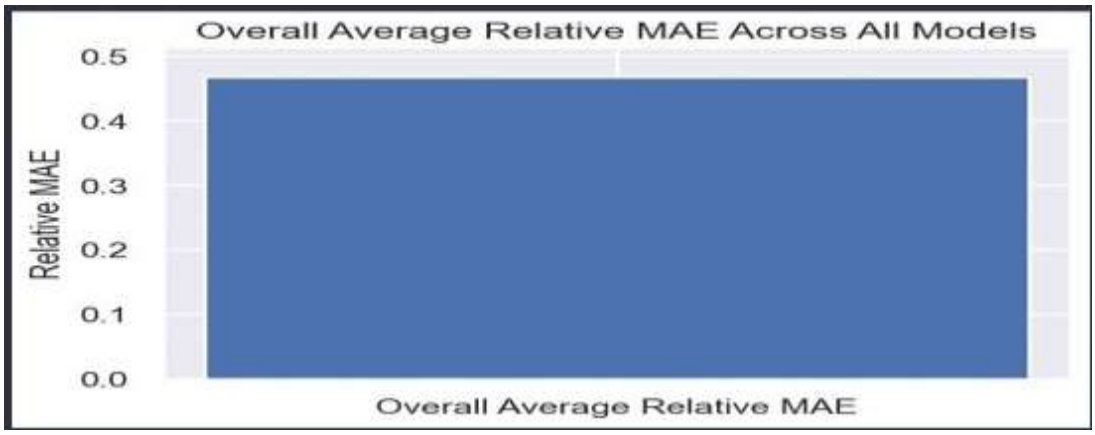

Figure 12

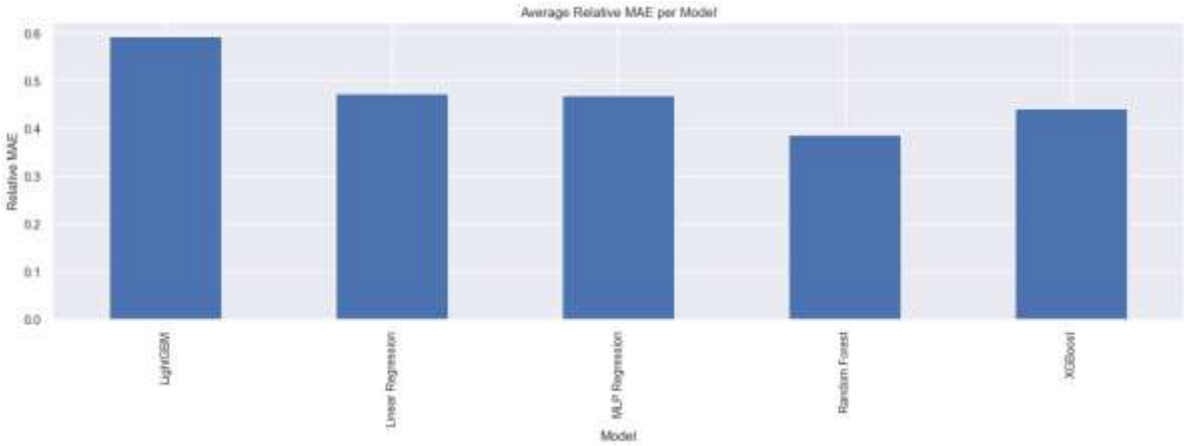

Figure 13

For detailed results, it is recommended to refer to the accompanying Excel sheet in the appendix, where a more comprehensive breakdown and analysis of the predictions are available.

	Store_Number	Product_Type	Model	MAE	Sales_Range	Average_Sales_Unit	Predicted_Value	Actual_Value
0	1	AUTOMOTIVE	Random Forest	2.19462897	19	3.25313943	[6.81333333 4.84166667 4.99440741	[8.0, 5.0, 4.0, 3.0, 8.0, 3.0, 6.0, 7.0, 4.0, 7.0, 9.0, 1.0, 6.0, 1.0, 1.0, 4.0]
1	1	BEAUTY	MLP Regression	1.63984209	12	2.40998275	[2.93550418 2.69332671 2.72649487	[3.0, 4.0, 2.0, 5.0, 7.0, 3.0, 2.0, 5.0, 2.0, 4.0, 10.0, 1.0, 3.0, 1.0, 6.0, 4.0]
2	1	BEVERAGES	Linear Regression	272.359973	5051	1588.692216	[2208.35486517 2412.04604528	[2414.0, 2627.0, 2645.0, 2037.0, 2479.0, 2093.0, 968.0, 2086.0, 2418.0, 2311.0, 2377.0, 1006.0, 1659.0, 803.0,
3	1	BOOKS	MLP Regression	0.27226108	8	0.689542484	[1.01626441 0.12563964 0.18408522	[1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
4	1	BREAD/BAKERY	MLP Regression	43.5420121	745.513	338.676297	[389.42160853 360.75771661	[370.994, 373.70898, 428.186, 282.83, 385.044, 271.579, 144.024, 328.382, 361.412, 399.547, 417.023, 145.0
5	1	CELEBRATION	Linear Regression	6.77698561	1306	12.72630781	[14.90087869 15.17114473 13.73200833	[3.0, 11.0, 7.0, 10.0, 24.0, 12.0, 2.0, 18.0, 11.0, 12.0, 17.0, 4.0, 3.0, 1.0, 4.0, 21.0]
6	1	CLEANING	XGBoost	101.44175	1859	640.8348188	[670.5358 680.39845 902.0665	[678.0, 867.0, 942.0, 622.0, 729.0, 442.0, 233.0, 741.0, 767.0, 653.0, 698.0, 341.0, 351.0, 369.0, 571.0, 703.0]
7	1	DAIRY	MLP Regression	71.3403043	1225	626.4729649	[691.55888801 669.85596702	[727.0, 888.0, 941.0, 529.0, 767.0, 624.0, 312.0, 717.0, 569.0, 789.0, 736.0, 343.0, 526.0, 268.0, 699.0, 902.0]
8	1	DELI	LightGBM	20.3003768	393.448	123.0168834	[160.80411541 140.62477338	[189.11, 133.588, 150.745, 106.929, 170.185, 104.188, 60.289, 138.907, 130.678, 123.498, 139.761, 84.303, 9
9	1	EGGS	MLP Regression	16.3363662	246	140.5264409	[161.20963879 122.16795968	[149.0, 161.0, 180.0, 120.0, 143.0, 155.0, 47.0, 152.0, 130.0, 134.0, 114.0, 86.0, 113.0, 60.0, 170.0, 131.0]
10	1	FROZEN FOODS	XGBoost	23.3318637	5608.718	129.4868592	[153.76108 109.71914 159.6527	[158.354, 144.126, 200.899, 132.185, 143.0, 155.597, 40.0, 136.719, 134.287, 154.309, 167.70999, 61.0, 107.7

Figure 14

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15. Appendix

Our code:

Code part 1

Code part 2

Results files:

Result excel file